CBSE Class 10 Question & Paper Generator — Build Prompt

Nov 5, 2026 · @Nikhil

How to use this doc

Paste the sections into your vibe-coding tool in order, one phase at a time, not all at once. Tools like Lovable, Bolt, v0, Cursor or Claude Code build far better from staged prompts than from one giant message.

1. **Phase 1 prompt** = the Master build prompt (next section) + the Subjects/chapters data + the Blueprints. This builds the UI, data model and paper templates with mock questions.
2. **Phase 2 prompt** = the AI question engine + Backend system prompts. This wires real generation through free-tier LLMs.
3. **Phase 3 prompt** = the Board-style PDF design spec. This makes the downloads look like a real board paper.
4. **Phase 4** = the Acceptance tests at the end. Paste them and ask the tool to run each one and fix failures.

Facts here were checked on 5 Nov 2026 against CBSE's 2026-27 sample papers (released 21 Sep 2026; CBSE states there is no change in question paper design for 2026-27). Before launch, verify every chapter list and blueprint against the official PDFs at cbseacademic.nic.in — the app is built so these live in editable config files, not hard-coded logic.

Two honest limits to keep in mind:

- No free LLM is "highly accurate" on its own. Accuracy comes from the pipeline in the engine section: generate → solve independently → verify with a second model → check maths in code → human-reviewable flags.
- Reference books and coaching modules (Oswaal, Arihant, RD Sharma, PW, Allen modules etc.) are copyrighted. The app must not scrape or copy them. It learns *style and difficulty* from freely published CBSE material (NCERT textbooks, NCERT Exemplar, CBSE sample papers, marking schemes, previous-year papers) and writes original questions.

Master build prompt (Phase 1 — paste this first)

```
ROLE
You are a senior full-stack engineer and an experienced CBSE Class 10 paper-setter. Build a production-quality web app (mobile-responsive, installable as a PWA) called "BoardReady 10" that generates CBSE Class 10 practice questions and full 80-mark sample papers as downloadable PDFs.

NON-NEGOTIABLE PRINCIPLES
1. Syllabus fidelity: only generate questions from the CBSE 2026-27 Class X syllabus and NCERT textbooks. Never include deleted/rationalised topics. Chapters flagged "internal assessment only" or "map only" must follow those flags.
2. Blueprint fidelity: full papers must match the official 2026-27 section structure, question count, marks per question, internal choices and total = 80 exactly. Blueprints live in /config/blueprints/*.json, never hard-coded.
3. Originality: every generated question must be new (see novelty check). Never copy text from copyrighted books or coaching modules.
4. Correctness: no question is shown to the user unless its answer has been independently solved and verified (see engine spec). Failed items are regenerated, not shown.
5. Config-driven: subjects, chapters, topics, levels and blueprints are JSON/YAML files an admin can edit without touching code.

TECH STACK (all free tiers)
- Frontend: Next.js (App Router) + TypeScript + Tailwind + shadcn/ui.
- Backend: Next.js API routes (or a small FastAPI service if the tool prefers Python for maths verification).
- Database/auth/storage: Supabase free tier (Postgres + pgvector for embeddings, Storage for PDFs, Auth with Google + email).
- PDF: HTML → PDF with Playwright/Chromium (server) using KaTeX for maths and SVG for diagrams. Fallback: @react-pdf/renderer.
- LLM access: an OpenAI-compatible router over free tiers (Groq, Google Gemini, OpenRouter :free models, NVIDIA NIM, Cerebras, Mistral). All API keys only in server env vars, never in the browser.
- Maths checking: SymPy (Python) or mathjs (Node) for numeric/symbolic verification.
- Deploy: Vercel (web) + Supabase; optional Render/Railway free instance for the Python worker.

APP STRUCTURE — HOME SCREEN
Two large cards: "Question Generator" and "Sample Paper Generator". Below: "My History" (past generations, re-download), "Question Bank" (saved/starred
```

```
questions), Settings (theme, default answer-key toggle, font size).

FLOW A — QUESTION GENERATOR (step wizard, one choice per screen, back button
always works)
Step 1 Subject: show every Class X subject from /config/subjects.json grouped
as: Core (Mathematics Standard 041, Mathematics Basic 241, Science 086,
Social Science 087), English (listed SEPARATELY as: English Language &
Literature 184 — Reading; English Grammar; English Writing Skills; English
Literature; English Communicative 101), Hindi (Hindi Course A 002, Hindi
Course B 085), Other languages, Skill subjects and Electives. Search bar on
top.
Step 2 Chapter: chapters for that subject in NCERT order, grouped by
book/unit (e.g. SST → History/Geography/Political Science/Economics; Science
→ Biology/Chemistry/Physics in board-paper order). Multi-select allowed +
"Select all". Show badges: "Map only", "Internal assessment only" (disabled
for board-style generation with a tooltip). For English Grammar and Writing
Skills show topics instead of chapters (see data section).
Step 3 Number of questions: stepper 1–50 (default 10). Optional filter chips
for question type: MCQ, Assertion-Reason, Very Short (2), Short (3), Long
(5), Case/Source-based (4), Map, Diagram-based, Numerical. Default = "Mixed
as per board".
Step 4 Level: Easy / Medium / Moderate / Hard / Very Hard, each with a one-
line description from /config/levels.json. Toggle: "Competency-based focus"
(on by default — at least 50% competency-based, matching CBSE's stated
weight).
Step 5 Output options: Include answer key (separate PDF / at end / none),
include marking-scheme style step marks, include hints, time-per-question
estimate.
Generate → progress screen showing live stages (Planning → Writing → Solving
→ Verifying → Formatting) → preview page with each question rendered (KaTeX)
+ answer collapsible → buttons: Download Question PDF, Download Answer Key
PDF, Regenerate one question, Replace with harder/easier, Star to bank,
Report error.

FLOW B — SAMPLE PAPER GENERATOR
Step 1 Subject (only subjects that have a blueprint file; others show
"blueprint coming soon").
Step 2 Level: Easy / Medium / Moderate / Hard / Very Hard (paper-level mix,
defined in /config/levels.json).
Step 3 Options: Set code (e.g. 30/1/1 style label you invent, never a real
CBSE code), include marking scheme, include internal choices (default ON, as
in board papers), "avoid questions I have already seen" (default ON).
Generate → build the paper strictly from the blueprint: for every slot
```

```
(section, Q no., type, marks, chapter weight, competency flag, internal
choice) request a question from the engine; run paper-level checks (total 80,
chapter weightage within ±2 marks of syllabus unit weightage, no duplicate
concepts, time-to-solve within 180 min for the level) → preview → download
Question Paper PDF + Marking Scheme PDF.

DATA MODEL (Postgres)
subjects(id, code, name, group, has_blueprint)
chapters(id, subject_id, unit, book, order, name,
assessment_flag[board|ia_only|map_only], weight_marks)
topics(id, chapter_id, name, learning_outcomes[])
questions(id, subject_id, chapter_id, topic_ids[], type, marks, level,
competency_based bool, bloom, stem, options jsonb, assertion, reason,
case_text, sub_parts jsonb, diagram_svg, answer, solution_steps jsonb,
marking_scheme jsonb, est_time_sec, source_style, embedding vector(768),
verified bool, verifier_notes, model_used, created_at)
generations(id, user_id, mode[questions|paper], params jsonb, question_ids[],
pdf_url, key_url, created_at)
reports(id, question_id, user_id, reason, created_at)
seen(user_id, question_id)

UX RULES
- Indian student audience: clean, fast, works on cheap Android phones and
slow 4G. Skeleton loaders, no heavy animations.
- Light/dark mode; Hindi UI strings ready (i18n) even if English ships first.
- Rate-limit per user (e.g. 5 papers/day on free plan) to stay inside free
LLM quotas; show a friendly quota message.
- Every generated PDF has the footer: "Practice paper generated by BoardReady
10 — not an official CBSE paper." Never use the CBSE logo.

BUILD NOW: scaffold the project, the wizard flows, the data model, the config
loaders, and a MOCK question engine that returns placeholder questions with
the correct shape, so the full flow and PDF download work end to end. Real AI
comes in Phase 2.
```

Subjects, chapters and levels (paste with Phase 1)

The core-subject chapter lists below follow the current rationalised NCERT books. Items marked (verify) are ones whose board-exam status you should confirm in the official 2026-27 syllabus PDF before launch. Languages and skill subjects are loaded from the syllabus PDFs rather than typed from memory.

```
# /config/subjects.yaml — seed data. Admin-editable.
mathematics_standard_041: &maths
  - Real Numbers
  - Polynomials
  - Pair of Linear Equations in Two Variables
  - Quadratic Equations
  - Arithmetic Progressions
  - Triangles
  - Coordinate Geometry
  - Introduction to Trigonometry
  - Some Applications of Trigonometry
  - Circles
  - Areas Related to Circles
  - Surface Areas and Volumes
  - Statistics
  - Probability
mathematics_basic_241: *maths   # same chapters, easier depth; different
blueprint difficulty

science_086:   # paper order: Biology → Chemistry → Physics
  biology: [Life Processes, Control and Coordination, How do Organisms
Reproduce?, Heredity, Our Environment]
  chemistry: [Chemical Reactions and Equations, Acids Bases and Salts, Metals
and Non-metals, Carbon and its Compounds]
  physics: [Light – Reflection and Refraction, The Human Eye and the
Colourful World, Electricity, Magnetic Effects of Electric Current]

social_science_087:
  history (India and the Contemporary World II):
    [The Rise of Nationalism in Europe, Nationalism in India, The Making of a
Global World (verify sub-topics), The Age of Industrialisation (verify),
Print Culture and the Modern World]
  geography (Contemporary India II):
    [Resources and Development, Forest and Wildlife Resources (verify: may be
IA-only), Water Resources, Agriculture, Minerals and Energy Resources,
Manufacturing Industries, Lifelines of National Economy (MAP ONLY in board)]
  political_science (Democratic Politics II):
    [Power-sharing, Federalism, Gender Religion and Caste, Political Parties,
Outcomes of Democracy]
  economics (Understanding Economic Development):
    [Development, Sectors of the Indian Economy, Money and Credit,
Globalisation and the Indian Economy, Consumer Rights (project work only –
not in board)]
```

```
  map_list: load verbatim from the syllabus PDF's map-work annexure (History
+ Geography items)

english_language_literature_184:
  reading: [Discursive passage, Case-based factual passage]
  grammar: [Tenses, Modals, Subject–verb concord, Reported speech (commands &
requests, statements, questions), Determiners]
  writing_skills: [Formal letter (complaint / enquiry / order / letter to the
editor), Analytical paragraph (from chart, graph, table, cue)]
  literature_first_flight_prose: [A Letter to God, Nelson Mandela: Long Walk
to Freedom, Two Stories about Flying, From the Diary of Anne Frank, Glimpses
of India, Mijbil the Otter, Madam Rides the Bus, The Sermon at Benares, The
Proposal]
  literature_first_flight_poems: [Dust of Snow, Fire and Ice, A Tiger in the
Zoo, How to Tell Wild Animals, The Ball Poem, Amanda!, The Trees, Fog, The
Tale of Custard the Dragon, For Anne Gregory]
  literature_footprints_without_feet: [A Triumph of Surgery, The Thief's
Story, The Midnight Visitor, A Question of Trust, Footprints without Feet,
The Making of a Scientist, The Necklace, The Book That Saved the Earth]   #
(verify list)
# In the UI, show English Grammar and English Writing Skills as their OWN
subject tiles
# (they map to 184 grammar/writing_skills above) so students can practise
them separately.

english_communicative_101: load from syllabus PDF (Interact in English books
+ grammar + writing)
hindi_course_a_002: load from syllabus PDF (Kshitij-2, Kritika-2 + vyakaran +
lekhan)
hindi_course_b_085: load from syllabus PDF (Sparsh-2, Sanchayan-2 + vyakaran
+ lekhan)
other_languages: Sanskrit, Urdu, Marathi, Bengali, Tamil, Telugu, Gujarati,
Kannada, Malayalam, Punjabi, French, German, Spanish, Japanese … (full list
from CBSE Class X scheme of studies)
skill_and_electives: Information Technology 402, Artificial Intelligence 417,
Computer Applications 165, Home Science, Painting, Music … (from scheme of
studies)
```

Difficulty levels (single questions)

Level	What it means	Thinking	Typical trap	Time per mark
Easy	Direct NCERT recall or one-step use of a formula/definition	Remember, Understand	None	~1 min
Medium	NCERT in-text / exercise level, 2–3 steps, familiar context	Understand, Apply	One common slip (units, sign)	~1.5 min
Moderate	NCERT Exemplar / CBSE sample-paper level; unfamiliar context, combines 2 ideas	Apply, Analyse	Distractors built from real misconceptions	~2 min
Hard	Board "toughest 10%" level; multi-concept, case/data interpretation, careful reading	Analyse, Evaluate	Hidden condition, extra data, near-miss options	~2.5 min
Very Hard	Olympiad-lite but strictly in syllabus; long chains, non-routine construction, proof or justification	Evaluate, Create	Two-layer trap; answer only after checking all cases	~3+ min

Paper-level mix (Sample Paper Generator)

Paper level	Easy	Medium	Moderate	Hard	Very Hard
Easy	40%	35%	20%	5%	0%
Medium	25%	35%	25%	15%	0%
Moderate (board-like)	20%	30%	30%	15%	5%
Hard	10%	20%	30%	30%	10%
Very Hard	5%	10%	25%	35%	25%

Percentages are of the 80 marks. In every level, at least 50% of marks must be competency-based (case-based, source-based, assertion-reason, data/graph, real-life application), matching CBSE's stated weight. Keep these numbers in /config/levels.json so they can be tuned after testing with real students.

Paper blueprints for 2026-27 (paste with Phase 1)

Every main paper is 80 marks in 3 hours; these structures come from the official 2026-27 sample papers as analysed by Competishun. Store each as /config/blueprints/<code>.json and confirm against the SQP PDFs on cbseacademic.nic.in.

Mathematics Standard 041 and Basic 241 — 38 questions

Section	Q no.	Type	Marks each	Total	Internal choice
A	1–20	18 MCQ + 2 Assertion-Reason (Q19, Q20)	1	20	None
B	21–25	Very short answer	2	10	In 2 questions
C	26–31	Short answer	3	18	In 2 questions
D	32–35	Long answer	5	20	In 2 questions
E	36–38	Case study, sub-parts 1 + 1 + 2	4	12	In the 2-mark sub-part

Rule: no calculators; take π = 22/7 unless stated. Standard = multi-step reasoning; Basic = same structure, more direct application.

Science 086 — 39 questions, three subject sections

Section	Q no.	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Total
A Biology	1–16	9	3	2	1	1	30
B Chemistry	17–29	8	1	2	1	1	25
C Physics	30–39	3	2	3	1	1	25

Rules: 1-mark sets include Assertion-Reason items in each section; 4-mark items are case/experiment based; answers written in the wrong section are not assessed, so the instructions page must say "Answer each section in its own part of the answer book".

Social Science 087 — 38 questions, four sections of 20 marks

Section	MCQ (1)	VSA (2)	SA (3)	LA (5)	Case (4)	Map	Total
A History	4	1	1	1	1	2 (Q9)	20
B Geography	6	1	0	1	1	3 (Q19)	20
C Political Science	4	2	1	1	1	—	20
D Economics	6	0	3	1	0	—	20

Word limits printed in instructions: VSA up to 40 words, SA up to 60, LA up to 120, case-based up to 100 (3 sub-questions). Map questions need an outline map of India page in the PDF, plus an alternative question for visually impaired candidates (standard CBSE practice).

English Language & Literature 184 — 11 questions

Section	Contents	Marks
A Reading	2 unseen passages × 10 (one discursive, one case-based/factual)	20
B Writing & Grammar	Grammar: attempt any 10 of 12 items (10); formal letter (5); analytical paragraph (5), each with choice	20
C Literature	2 extracts (5 + 5); short answers 4 × 3 and 2 × 3; two long answers × 6	40

English Communicative 101 — 12 questions

Section	Contents	Marks
A Reading	2 passages (12 + 10)	22
B Writing	Application (3), factual description (4), letter to the editor (7), article (8)	22
C Grammar	Gap filling (3), omission (4), do as directed (3)	10
D Literature	Extracts 2 × 4, short answers 5 × 2, one long answer × 8	26

Everything else

Hindi A/B, other languages and skill subjects get blueprints by having the admin upload that subject's official SQP PDF; the app's "Blueprint extractor" (an LLM call with a strict JSON schema) drafts the JSON, and an admin approves it before the subject appears in the Sample Paper Generator.

Blueprint JSON shape for every subject:

```
{
  "subject_code": "041", "session": "2026-27", "max_marks": 80,
"duration_min": 180,
  "general_instructions": ["This question paper contains 38 questions. All
questions are compulsory.", "..."],
  "sections": [
    {"id": "A", "title": "Section A", "note": "Section A consists of 20
questions of 1 mark each.",
     "slots": [
       {"q": 1, "type": "MCQ", "marks": 1, "internal_choice": false},
       {"q": 19, "type": "ASSERTION_REASON", "marks": 1, "internal_choice":
false}
     ]}
  ],
  "chapter_weightage": {"Real Numbers": 6, "...": 0},
  "competency_min_pct": 50
}
```

Chapter/unit weightage (e.g. Maths unit marks, Science unit marks) comes from the official syllabus PDF and must be filled in before the paper generator goes live.

AI question engine (Phase 2 — paste this second)

The engine is a separate backend module that never trusts one model: a question reaches the user only after a different model has solved it blind and agreed with the key. Free-tier quotas as of Oct 2026: Groq about 1,000 requests/day, OpenRouter :free models 50/day without credit, NVIDIA's API catalog free for Developer Program members, plus Gemini, Mistral and Cerebras free tiers (source, source). These limits change often, so they live in config.

```
BUILD THE QUESTION ENGINE (/engine)

1. PROVIDER ROUTER (/engine/router.ts)
- One OpenAI-compatible client; providers defined in /config/providers.yaml:
  name, base_url, env_key, models{generator, solver, verifier, embedder},
rpm, rpd, tpm, priority, enabled.
- Seed providers: Groq, Google Gemini (OpenAI-compatible endpoint),
OpenRouter (:free models, e.g. DeepSeek/Qwen/Llama/Gemma when listed), NVIDIA
NIM (build.nvidia.com), Cerebras, Mistral. DeepSeek's own API is paid (cheap)
— add as optional low-cost fallback.
- Each role uses a DIFFERENT provider/model family where possible (generator
≠ solver ≠ verifier) to avoid shared blind spots.
- Token-bucket rate limiter per provider (RPM + RPD) stored in
Postgres/Redis; on 429/5xx/timeout (20 s) → exponential backoff once, then
fail over to next provider by priority.
- Track daily usage per provider; admin dashboard shows remaining quota.
- All prompts request JSON with a schema; validate with Zod; on invalid JSON
→ one repair retry, then fail over.

2. KNOWLEDGE BASE (RAG) (/engine/kb)
- Ingest ONLY freely published sources: NCERT Class 10 textbooks
(ncert.nic.in), NCERT Exemplar, CBSE 2026-27 syllabus, CBSE sample papers +
marking schemes (current and past years), CBSE previous-year board papers,
CBSE competency-based question banks/learning-outcome documents.
- Chunk by chapter → topic; store embeddings in pgvector with metadata
{subject, chapter, topic, source_type, year, marks, type}.
- Purpose: (a) ground every question in syllabus facts, (b) give the
generator 3–5 STYLE EXEMPLARS of the requested type/marks/level, (c) build
the novelty index of existing questions.
- Copyrighted reference books / coaching modules are NOT ingested. If an
admin uploads their own notes, store them as private style hints only, never
as text to reproduce.
```

```
3. GENERATION PIPELINE (per requested question)
  a. PLAN: pick topic(s) inside the chosen chapter(s) not recently used for this user; pick type, marks, level, competency flag, Bloom level, and a "twist" from /config/twists.yaml (real-life Indian context, data table, graph reading, reverse problem, multi-concept link, error-spotting, what-if change, assertion built on a common misconception).
  b. RETRIEVE: top-k syllabus chunks for the topic + 3 exemplar questions of same type/marks/level (exemplars are for style only).
  c. GENERATE (generator model, temperature 0.8): returns the full question JSON (stem, options, answer, step solution, marking scheme, est_time, misconception each distractor targets).
  d. SOLVE BLIND (solver model, temperature 0): receives the question WITHOUT the answer; returns its own answer + steps.
  e. VERIFY (verifier model, temperature 0): compares key vs blind solution, checks syllabus scope, single correct option, no ambiguity, grammar, marks–effort fit, level fit, word limits; returns pass/fail + issues.
  f. COMPUTE-CHECK (code, not LLM): for Maths/Physics/Chemistry numericals, re-evaluate the final answer with SymPy/mathjs from a machine-readable expression the generator must include; balance chemical equations programmatically; for MCQs ensure exactly one option equals the computed answer.
  g. NOVELTY: embed the stem; reject if cosine similarity ≥ 0.88 to any question in the KB (sample papers, PYQs, exemplar) or ≥ 0.85 to anything this user has seen. Also reject if a 12-word span matches any KB text verbatim.
  h. If any check fails → feed the issues back to the generator for one revision; still failing → discard and re-plan with a different twist. Max 3 cycles per slot, then mark slot "needs review" and use a verified question from the bank.
  i. SAVE to questions table with verified=true, model names, verifier notes.

4. BATCH + SPEED
- Generate a paper's slots in parallel (respecting rate limits), cheap slots (1-mark) on faster models.
- Pre-warm: a nightly scheduled job generates and verifies a buffer of questions per chapter × type × level, so most user requests are served instantly from fresh, never-seen, verified stock and the live LLM is used only to top up.

5. QUALITY LOOP
- "Report error" from users → question hidden after 2 reports pending admin review.
```

```
- Admin review screen: approve / edit / reject, with diff of key vs blind
solution.
- Log per-provider pass rates; auto-lower priority of models whose verify-
pass rate drops below 70%.
```

Backend system prompts (Phase 2 — store in /engine/prompts)

These four prompts are what the app sends to the free LLMs; the vibe-coding tool should save them as files and fill the {placeholders} at runtime.

1. Generator

```
You are a senior CBSE Class 10 paper-setter with 20 years of experience
setting board papers and marking schemes. You write ORIGINAL questions for
the CBSE {session} Class X syllabus.

TASK
Write {n} question(s):
- Subject: {subject} ({code}); Chapter(s): {chapters}; Topic(s): {topics}
- Type: {type}; Marks: {marks}; Level: {level} = {level_definition}
- Competency-based: {true/false}; Bloom level: {bloom}; Twist: {twist}

GROUNDING (syllabus facts you may use — do not go beyond them):
{syllabus_chunks}

STYLE EXEMPLARS (match format, tone and difficulty; NEVER reuse their
wording, numbers, names or scenarios):
{exemplars}

HARD RULES
1. Stay strictly inside the rationalised NCERT/CBSE {session} syllabus. No
deleted topics, no Class 11 concepts, no formulas students are not taught.
2. Original: new scenario, new numbers, new names (use Indian contexts:
cities, festivals, sports, ISRO, railways, farming, markets, UPI, local
weather).
3. Board language: crisp, unambiguous, the way CBSE phrases questions
("Justify your answer.", "Give reason.", "Show that…", "Prove that…").
4. Marks must match effort: 1 mark = one idea; 2 marks = two points/steps; 3
marks = three; 5 marks = full derivation/explanation; case-based 4 = sub-
parts 1+1+2 (Maths/Science) or as per blueprint.
5. MCQ: exactly 4 options (a)–(d), ONE correct; every distractor targets a
```

```
named common misconception; no "all of the above"; options similar length;
avoid giveaway words.
6. Assertion-Reason: use the exact CBSE option set:
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true but R is not the correct explanation of A.
   (c) A is true but R is false.
   (d) A is false but R is true.
7. Case/source-based: 120–200 word passage or a data table/graph described in
text; sub-questions must need the passage.
8. Numericals: include "verify_expr" — a SymPy-parsable expression whose
value equals the final answer; use π = 22/7 unless stated; SI units; sensible
values a student can compute without a calculator.
9. Diagrams: if needed, return clean SVG (black lines, labelled, no colour
reliance) in "diagram_svg".
10. Difficulty for {level}: {level_rules}. Hard/Very Hard must be hard
because of thinking (multi-concept link, hidden condition, data
interpretation), NOT because of obscure facts or long arithmetic.
11. Internal choice (if {internal_choice}=true): give an OR question of equal
marks, equal difficulty, different topic/skill.

OUTPUT: JSON only, matching this schema:
{"questions":
[{"type","marks","level","competency_based","bloom","chapter","topics":
[],"stem","options":
{"a","b","c","d"}|null,"assertion"|null,"reason"|null,"case_text"|null,"sub_p
arts":
[{"label","text","marks"}]|null,"or_question"|null,"diagram_svg"|null,"answer
","solution_steps":[],"marking_scheme":
[{"step","marks"}],"verify_expr"|null,"distractor_rationale":
{},"est_time_sec","word_limit"|null}]}
Use LaTeX inside $...$ for all maths.
```

2. Blind solver

```
You are a top CBSE Class 10 student who scores 100/100. Solve the question
below exactly as you would in the board exam. You do NOT see any answer key.
- Show every step a board examiner would award marks for.
- For MCQ/Assertion-Reason, give the option letter AND a one-line
justification for why each other option is wrong.
- If the question is ambiguous, has no correct option, more than one correct
option, missing data, or uses something outside the Class 10 syllabus, say so
in "problems".
QUESTION: {question_without_answer}
OUTPUT JSON: {"final_answer", "steps":[], "time_taken_estimate_sec",
"problems":[]}
```

3. Verifier

```
You are the CBSE Head Examiner reviewing a new question before it is printed.
Be strict; a wrong question in a board-style paper is unacceptable.
INPUTS: question + setter's key + setter's marking scheme:
{question_with_key}; independent blind solution: {blind_solution}; syllabus
scope: {syllabus_chunks}; requested level: {level} ({level_definition});
requested marks: {marks}.
CHECK EACH and return pass/fail per item:
1 answer_matches (key and blind solution agree; if not, decide who is right
and why)
2 single_correct_answer (MCQ/A-R)
3 in_syllabus (no deleted/out-of-scope content)
4 unambiguous (one reading only; all data given; units consistent)
5 marks_effort_fit
6 level_fit (would a typical student find this {level}?)
7 competency_fit (if competency_based=true, does it test
application/analysis, not recall?)
8 language_quality (grammar, CBSE phrasing, word limits)
9 marking_scheme_valid (step marks add up to total)
10 originality (does it look copied from a well-known textbook exercise? flag
if so)
OUTPUT JSON: {"verdict":"pass"|"revise"|"reject", "checks":{...}, "issues":
[], "suggested_fix": "..."}
verdict=pass only if ALL checks pass.
```

4. Paper assembler (final check over the whole paper)

```
You are moderating a complete CBSE Class 10 {subject} sample paper for
session {session}, level {paper_level}.
INPUTS: blueprint {blueprint_json}; assembled paper {paper_json}; chapter
weightage {weightage}.
VERIFY:
- Question count, numbering, section order, marks per question and total =
{max_marks} exactly match the blueprint.
- Internal choices exist exactly where the blueprint says, and each OR pair
is equal in marks and difficulty.
- Chapter weightage within ±2 marks of target; no chapter missing that the
blueprint requires.
- No two questions test the same concept the same way; no question gives away
another's answer.
- Difficulty mix within ±5% of the {paper_level} targets; ≥50% marks
competency-based.
- Estimated total writing + thinking time fits 180 minutes for an average
student at this level (allow 15 min reading).
- General instructions text matches the blueprint's instructions.
OUTPUT JSON: {"verdict":"pass"|"fix", "problems":
[{"q":n,"issue","action":"replace"|"adjust"}]}
```

The app replaces any question the assembler flags, re-runs the assembler, and only then renders the PDF.

Board-style PDF design spec (Phase 3)

The PDF should look and feel like a real board paper in layout and typography, while clearly labelled as a practice paper and free of CBSE's logo or real paper codes.

```
BUILD THE PDF RENDERER (/pdf) — HTML templates → Playwright Chromium → PDF

PAGE
- A4 portrait, margins 18 mm top/bottom, 20 mm left/right. Single column.
Black text on white.
- Fonts: "Times New Roman"-style serif for body (embed Tinos or Liberation
Serif, metric-compatible and free), 12 pt, line-height 1.35. Bold sans
(Arimo/Liberation Sans) for section headers. Devanagari: Noto Serif
Devanagari for Hindi papers. Maths: KaTeX with its own fonts, server-side
rendered.
- Footer every page: centred page number "Page X of Y" ; right: "P.T.O." on
```

```
every page except the last; left tiny grey: "Practice paper – BoardReady 10 – not an official CBSE paper".

PAGE 1 HEADER (mimic layout, not identity)
- Top-left: "Series: XYZW/1" (invented). Top-right: "SET ~1" and "Q.P. Code: 30/1/1-P" (invented, always ending -P for Practice).
- A row of 8 empty bordered boxes labelled "Roll No." and a line "Candidates must write the Q.P. Code on the title page of the answer-book."
- Centered: subject name in caps + code, e.g. "MATHEMATICS (STANDARD) – 041" and "Practice Paper 2026-27 · Level: Moderate".
- Line: "Time allowed: 3 hours" (left) and "Maximum Marks: 80" (right), with a horizontal rule.
- A boxed note list (as on real papers): "Please check that this question paper contains __ printed pages", "__ questions", "15 minute time has been allotted to read this question paper" etc., filled automatically from the generated paper.

GENERAL INSTRUCTIONS
- Heading "General Instructions:" then a numbered list from blueprint.general_instructions (e.g. "This question paper contains 38 questions. All questions are compulsory." / "Section A consists of 20 questions of 1 mark each." …) — auto-filled with real counts.

BODY
- Section banner: centred bold "SECTION A" then italic line "This section consists of 20 questions of 1 mark each."
- Question row: question number right-aligned in a fixed 10 mm left column ("1."), text in the middle, marks in a fixed 10 mm right column ("1", "2"…). Use a CSS grid so marks never wrap.
- MCQ options: 2×2 grid when all options ≤ 30 characters, otherwise 4 lines; labels (a) (b) (c) (d).
- Assertion-Reason: "Assertion (A):" and "Reason (R):" in bold labels, then the four standard options.
- Internal choice: centred bold "OR" between the two alternatives.
- Case-based: boxed passage with light 0.5 pt border, then sub-parts (i) (ii) (iii) with marks per sub-part.
- Diagrams: inline SVG centred, max width 70%, labelled "Fig. 1" etc.
- Map questions (SST): separate last page "Outline map of India (Political)" with a public-domain outline SVG and the map instructions; plus a "For Visually Impaired Candidates only" alternative question.
- Never split a question across pages if it is shorter than 1/3 page (CSS break-inside: avoid); case passages may split only between sub-parts.
- Last line of paper: centred "● ● ●" or "*****" as end marker.
```

```
ANSWER KEY / MARKING SCHEME PDF
- Same header style titled "MARKING SCHEME – Practice Paper".
- Table per question: Q no. | Value points / steps | Marks (step-wise, e.g. "½ + ½", "1"). For MCQ: correct option + one-line reason. Total per section, grand total 80.

QUESTION-GENERATOR PDF (Flow A)
- Lighter header: subject, chapter(s), level, count, date. Numbered questions with marks; answer key on separate pages or separate file as chosen.

QA FOR THE RENDERER
- Snapshot test: generate a fixed seed paper for each blueprint and compare page count and layout across runs.
- Check: no overflowing maths, no orphan "OR", marks column aligned on every page, fonts embedded (pdffonts shows all embedded).
```

Acceptance tests and guardrails (Phase 4)

Paste this after the build and ask the tool to run every test, fix failures, and report results in a table.

```
RUN AND FIX UNTIL ALL PASS

FUNCTIONAL
1. Question Generator: Science → Electricity → 10 questions → Hard → PDF downloads, 10 questions, all with marks, answer key in separate PDF.
2. English Grammar tile → Reported speech → 15 questions → Medium → only grammar items, no literature.
3. English Writing Skills tile → Analytical paragraph → 3 → Moderate → each has a chart/table cue and word limit 100–120 words.
4. SST → Lifelines of National Economy → app shows "Map only" and generates map-type items only.
5. Sample Paper: Maths Standard → Moderate → 38 questions, sections A–E exactly as blueprint, total 80, OR in the right slots.
6. Sample Paper: Science → Hard → 39 questions, Biology 30 / Chemistry 25 / Physics 25.
7. Sample Paper: SST → Easy → 4 sections × 20, map Q9 (2) and Q19 (3), outline map page present.
8. Sample Paper: English 184 → Very Hard → 11 questions, reading 20 / writing+grammar 20 / literature 40.
9. Generating the same request twice gives different questions (novelty), and
```

```
no question the same user saw before.

CORRECTNESS (run on 100 generated items per core subject)
10. ≥ 98% pass blind-solve agreement; 100% of numericals pass the code check;
0 MCQs with two correct options.
11. 0 questions from deleted chapters (keyword guard list per subject +
verifier check).
12. Marking-scheme step marks always sum to the question's marks; paper
totals always 80.

RESILIENCE
13. Disable the top provider's key → generation still succeeds via fallback.
14. Simulate 429s → backoff + failover, user sees progress, not an error.
15. All providers down → serve verified questions from the pre-generated
bank; tell the user it's from the bank.

PDF
16. Fonts embedded; marks column aligned; no question split badly; footer
disclaimer on every page; no CBSE logo; Q.P. codes end in "-P".

SECURITY & COST
17. No API key visible in client bundle or network tab.
18. Per-user daily limits enforced; admin dashboard shows quota usage per
provider.
```

Guardrails worth keeping even if the tool pushes back:

- Keep the answer-verification step. It roughly triples LLM calls per question, which is why the nightly pre-generation buffer matters on free tiers.
- Keep the "not an official CBSE paper" label and invented paper codes. A PDF that could pass as a leaked or official board paper can mislead students and get the app taken down.
- Have a teacher or a strong Class 10 student review the first 50–100 generated questions per subject before you share the app widely; free models still make subtle syllabus and language slips that automated checks miss.

Sources

- CBSE Class X Sample Question Papers 2026-27 (official)
- Competishun — CBSE Sample Paper Class 10 2026-27 pattern analysis
- Top 8 free AI APIs, October 2026 (DEV Community)

- OpenRouter — Free LLM APIs compared